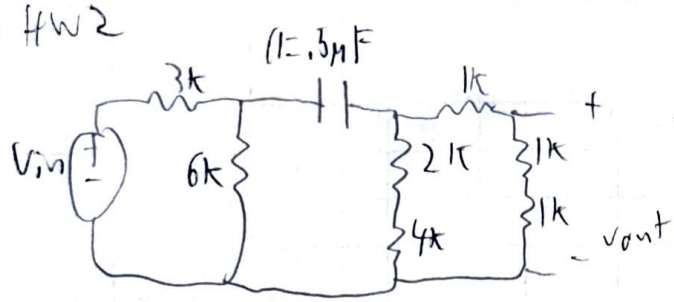
Elen 115 HW2

Dylan Trumbly

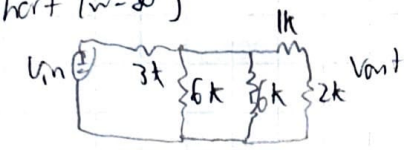


Vout is across both 1k's

open (w=0)

$\frac{V_{out}}{V_{in}} = 0 = \text{gain}$

short (w=∞)



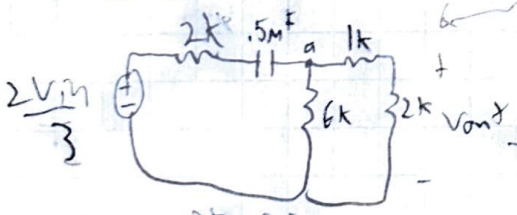
$\frac{V_{in}}{2}$

$V_{out} = \frac{V_{in}}{2}\left(\frac{2}{2+2.5}\right) = \frac{V_{in}}{4.5}$

$\frac{V_{out}}{V_{in}} = \frac{2}{9} = \text{gain}$

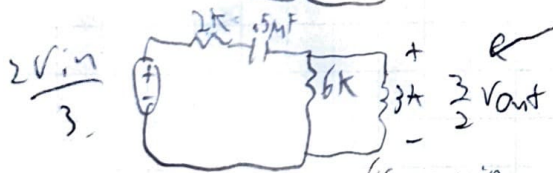
a. STC high pass w/ gain of $\frac{2}{9}$

therefore on left part

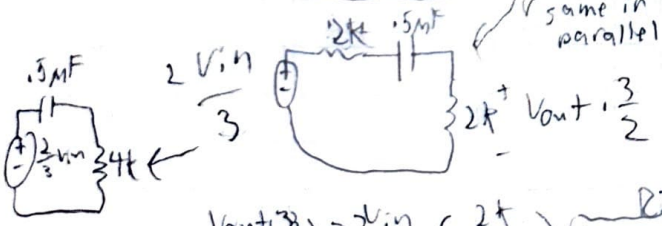


$V_{out} = \frac{2}{3}V_a$

$V_{out}\frac{3}{2} = a$



same in parallel



$V_{out}\left(\frac{3}{2}\right) = \frac{2V_{in}}{3}\left(\frac{2k}{4k}\right)$

$V_{out} = \frac{4}{9}V_{in}\left(\frac{1}{2}\right)$

$V_{out} = \frac{2}{9}V_{in}$ ✓

matches gain

R = 4k

b. $\frac{1}{2\pi 10^{-6} \cdot .5 \cdot 4k} = 79.577$ Hz

$w = 2\pi f$

$V_{in} = 3 + \sin(120\pi t) + 9\sin 2\pi 10^{10} t$

f=0	f=60	$f=10^{10}$
ignore	ignore	passes ✓

$V_{out} = \frac{2}{9}V_{in}$

$V_{out} = \frac{2}{9}(9\sin(2\pi 10^{10} t))$

$V_{out} = 2\sin(2\pi 10^{10} t)$

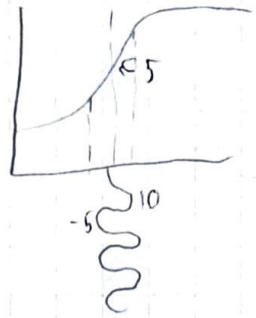
2. extend >10 and <-5
gain = 5

a.

i. 5V
ii. 1V

b. $2.5 + 7.5\sin\omega t$

bias point = 2.5
input V = 7.5V

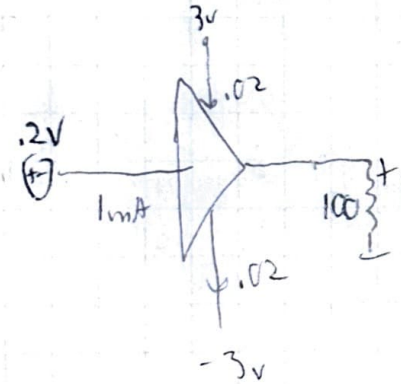


can't be >5
b/c of rules

$\frac{5}{5} = 1$
gain

3.

± 3V 2.2V peak over 100Ω load



$\frac{2.2}{100} = 22$ mA

a. Voltage gain $= \frac{2.2}{.2} = 11 \frac{V}{V} \rightarrow 20\log(11) = 20.828$ dB

current gain $= \frac{22}{1} = 22 \frac{A}{A} \rightarrow 20\log(22) = 26.848$ dB

power gain $= \frac{22(2.2)}{1(.2)} = 242 \frac{W}{W} \rightarrow 10\log(242) = 23.838$

b.

$3V \cdot .02A + -3V \cdot -.02A = .12$ mW

$120\text{mW} + \frac{\overset{\text{prms}}{.2(.001)}}{2} - \frac{\overset{\text{prms}}{2.2(.022)}}{2} = 95.9$ mW

$\frac{\frac{2.2(.022)}{2}}{.12} = .2016$

supply power = 120 mW
amp dissip. = 95.9 mW
amp eff. = 20.16%